

13. If  $F(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$ , show that  $F(x) F(y) = F(x + y)$ .

14. Show that

(i)  $\begin{bmatrix} 5 & -1 \\ 6 & 7 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix} \neq \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 & -1 \\ 6 & 7 \end{bmatrix}$

(ii)  $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 2 & 3 & 4 \end{bmatrix} \neq \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$

15. Find  $A^2 - 5A + 6I$ , if  $A = \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix}$

16. If  $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$ , prove that  $A^3 - 6A^2 + 7A + 2I = 0$

17. If  $A = \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix}$  and  $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ , find  $k$  so that  $A^2 = kA - 2I$

18. If  $A = \begin{bmatrix} 0 & -\tan \frac{\alpha}{2} \\ \tan \frac{\alpha}{2} & 0 \end{bmatrix}$  and  $I$  is the identity matrix of order 2, show that

$$I + A = (I - A) \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$$

19. A trust fund has ₹30,000 that must be invested in two different types of bonds. The first bond pays 5% interest per year, and the second bond pays 7% interest per year. Using matrix multiplication, determine how to divide ₹30,000 among the two types of bonds. If the trust fund must obtain an annual total interest of:

(a) ₹1800

(b) ₹2000

20. The bookshop of a particular school has 10 dozen chemistry books, 8 dozen physics books, 10 dozen economics books. Their selling prices are ₹80, ₹60 and ₹40 each respectively. Find the total amount the bookshop will receive from selling all the books using matrix algebra.

Assume X, Y, Z, W and P are matrices of order  $2 \times n$ ,  $3 \times k$ ,  $2 \times p$ ,  $n \times 3$  and  $p \times k$ , respectively. Choose the correct answer in Exercises 21 and 22.

21. The restriction on  $n$ ,  $k$  and  $p$  so that  $PY + WY$  will be defined are:

- (A)  $k = 3$ ,  $p = n$  (B)  $k$  is arbitrary,  $p = 2$   
(C)  $p$  is arbitrary,  $k = 3$  (D)  $k = 2$ ,  $p = 3$

22. If  $n = p$ , then the order of the matrix  $7X - 5Z$  is:

- (A)  $p \times 2$  (B)  $2 \times n$  (C)  $n \times 3$  (D)  $p \times n$

### 3.5. Transpose of a Matrix

In this section, we shall learn about transpose of a matrix and special types of matrices such as symmetric and skew symmetric matrices.

**Definition 3** If  $A = [a_{ij}]$  be an  $m \times n$  matrix, then the matrix obtained by interchanging the rows and columns of  $A$  is called the *transpose* of  $A$ . Transpose of the matrix  $A$  is denoted by  $A'$  or  $(A^T)$ . In other words, if  $A = [a_{ij}]_{m \times n}$ , then  $A' = [a_{ji}]_{n \times m}$ . For example,

$$\text{if } A = \begin{bmatrix} 3 & 5 \\ \sqrt{3} & 1 \\ 0 & -\frac{1}{5} \end{bmatrix}_{3 \times 2}, \text{ then } A' = \begin{bmatrix} 3 & \sqrt{3} & 0 \\ 5 & 1 & -\frac{1}{5} \end{bmatrix}_{2 \times 3}$$

#### 3.5.1 Properties of transpose of the matrices

We now state the following properties of transpose of matrices without proof. These may be verified by taking suitable examples.

For any matrices  $A$  and  $B$  of suitable orders, we have

- (i)  $(A')' = A$ , (ii)  $(kA)' = kA'$  (where  $k$  is any constant)  
(iii)  $(A + B)' = A' + B'$  (iv)  $(A B)' = B' A'$

**Example 20** If  $A = \begin{bmatrix} 3 & \sqrt{3} & 2 \\ 4 & 2 & 0 \end{bmatrix}$  and  $B = \begin{bmatrix} 2 & -1 & 2 \\ 1 & 2 & 4 \end{bmatrix}$ , verify that

- (i)  $(A')' = A$ , (ii)  $(A + B)' = A' + B'$ ,  
(iii)  $(kB)' = kB'$ , where  $k$  is any constant.